



Dynamics of dimeric kinesins: Limping, effect of longitudinal force, effects of neck linker extension and mutation, and comparison between kinesin-1 and kinesin-2



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ABSTRACT

Conventional kinesin (kinesin-1) can walk on microtubule filaments in an asymmetric hand-over-hand manner, exhibiting a marked alternation in the mean dwell time in successive steps. Here, we study computationally the asymmetric stepping dynamics of the kinesin-1 homodimer, revealing its origin and providing quantitative explanations of the available experimental data. The alternation in the mean dwell time in successive steps arises from the alternation in the mechanochemical coupling ratio, which is in turn caused by the alternation in the slight variation of the stretched neck linker length. Both the vertical and backward longitudinal forces can enhance the asymmetric ratio. Additionally, other aspects of the stepping dynamics of the dimer such as the velocity versus longitudinal force, extended neck linker, etc., are also studied. In particular, the conflicting experimental data, with some showing that the velocity does not change with the forward longitudinal load while others showing that the velocity increases largely with the forward longitudinal load, are explained quantitatively and consistently. The intriguing experimental data showing that cysteine-light *Drosophila* and human kinesin-1 mutants have different load-dependent velocity from the wild-type cases as well as that kinesin-2 dimers have different load-dependent velocity from the kinesin-1 are also explained consistently and quantitatively.

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1. Introduction

Conventional kinesin (kinesin-1) consists of two identical polypeptide subunits that dimerize to form a coiled-coil stalk. It is a molecular motor that can convert chemical energy of ATP hydrolysis to mechanical force. It plays essential roles in transporting membrane-bound vesicles and organelles in various cells by moving unidirectionally to the plus end of microtubule (MT) along paths parallel to the MT protofilaments for hundreds of 8-nm steps without dissociation [1–6].

Since its discovery, the mechanochemical properties of kinesin-1 have been studied extensively by using various experimental methods [1,4,5,7–10]. In particular, by using single-molecule optical trappings, diverse aspects of its moving dynamics have been studied elaborately [11–25]. While some single-molecule data showed that the movement velocity of the dimer versus the longitudinal load (the load along or opposite to the moving direction)

acting on the bead attached to the coiled-coil stalk has a sigmoid form, i.e., the velocity decreases with the increase of the backward load whereas does not increase with the increase of the forward load [20], other single-molecule data showed that the velocity versus load has a double-sigmoid form, i.e., the velocity decreases with the increase of the backward load and increases with the increase of the forward load [13,24]. However, the conflicting experimental data on the velocity versus the longitudinal load has not been explained reasonably up to now. More interestingly, by using single-molecule optical trappings with a micrometer bead attached to the coiled-coil stalk it was demonstrated that the homodimer moves in an asymmetric or limping hand-over-hand manner, i.e., the homodimer exhibits a marked alternation in the mean dwell time in successive steps [21,23]. To understand mechanism of the asymmetric or limping stepping, several models have been proposed [21,26]. Nevertheless, the molecular mechanism of the asymmetric or limping stepping of the homodimer is still undetermined. Although vertical force acting on the kinesin increases the limping effect [27], the molecular mechanism of how the vertical force increases the limping effect is unknown. Additionally, recent single-molecule studies of kinesin-2 motors such

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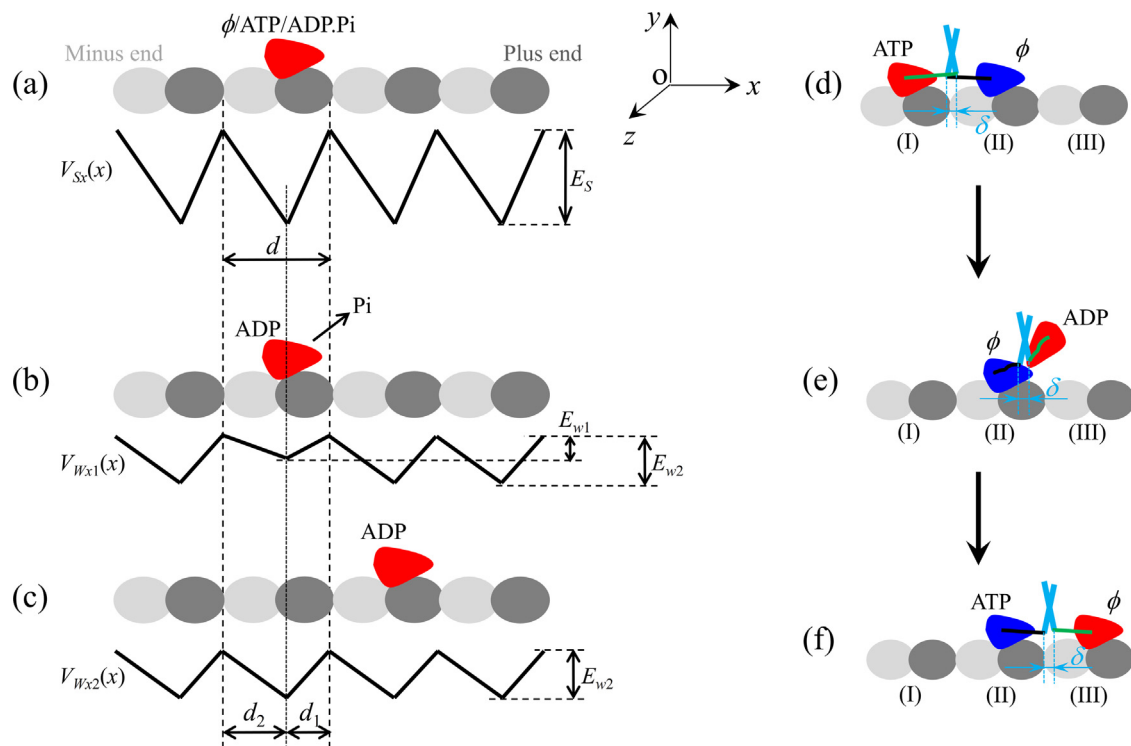


Fig. 1. Model of stepping of dimeric kinesin. Left panels illustrate interaction potentials between the single head and MT in the x direction (along a MT protofilament) during one ATPase cycle. (a) Strong interaction potential, $V_{Sx}(x)$, in nucleotide-free, ATP or ADP.Pi states. The potential depth E_S may be slightly different for different nucleotide states. The top figure shows the position of the kinesin head relative to the MT. (b) Weak interaction potential, $V_{Wx1}(x)$, in ADP state immediately after Pi release. The top figure shows the most probable position of the kinesin head relative to the MT. (c) Weak interaction potential, $V_{Wx2}(x)$, in ADP state in a period of time t_r after Pi release. The top figure shows the most probable position of the kinesin head relative to the MT. Right panels illustrate a forward step of the dimer. (d) Before stepping the trailing head (red) in ATP state binds to the rear MT-binding site (I) and the leading head (blue) in nucleotide-free state binds to the forward MT-binding site (II). (e) After ATP hydrolysis and then Pi release, due to the very weak affinity between the ADP-head and the local binding site (I) (see Fig. 1b), the trailing ADP-head (red) diffuses rapidly to the intermediate position relative to the other head (blue) due to the strong interaction between the two heads. (f) After ATP binding to the MT-bound head (blue) its neck linker docks, weakening the interaction between the two heads. Then the unbound head diffuses rapidly to the forward MT-binding site (III). Note that the neck linker docking in the MT-bound head provides an energy barrier to prevent the tethered ADP-head from moving backward to the rear MT-binding site (I) rather than acts as a driving force for the forward movement of the ADP-head. Note also that due to the presence of a small biased distance δ ($\ll d$) between the C-terminus end of one neck linker and that of another neck linker along the x direction, the length of the stretched linkers in the state with the two heads being bound simultaneously to MT alternates between two values in successive steps (d and f) (see Section 2.2 for detailed description). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

as heterodimeric KIF3A/B and homodimeric mutants KIF3A/A and KIF3B/B showed that the kinesin-2 motor velocity is less sensitive than kinesin-1 to external backward forces [28]. More puzzling, by swapping the neck-linker domains between kinesin-1 and kinesin-2, it was revealed that specific docking of the neck linker to the core motor domain is necessary for kinesin-1 but not for kinesin-2 [28]. Does kinesin-2 show different movement mechanism from kinesin-1? If the two motors share the same mechanism, why is the neck linker docking necessary for kinesin-1 but not for kinesin-2?

Recently, based on the previously proposed model [29,30] an improved model for the processive movement of dimeric kinesins was presented [31]. With the improved model diverse aspects of movement dynamics of the dimeric kinesin-1 were studied in detail, such as the effects of varying solution viscosity, varying the longitudinal load acting on the bead attached to the coiled-coil stalk and on the bead attached to one head of the dimer, nullifying the neck linker docking, extending the neck linkers, etc., on the movement dynamics. The theoretical data provided consistent and quantitative explanations of a lot of experimental data available in the literature. In this work, to understand the molecular mechanism of the asymmetric or limping stepping, with the improved model we provide quantitative explanations of the experimental data on the asymmetric stepping dynamics. Additionally, the conflicting experimental data on the velocity versus the forward longitudinal load are explained quantitatively. The puzzling experimental data, indicating that the specific docking of the neck linker to the core

motor domain is necessary for kinesin-1 but not for kinesin-2, are also explained quantitatively and well.

2. Methods

2.1. The model

First, we describe our model of processive movement of dimeric kinesin (Fig. 1) [31], which is modified from the model proposed before [29,30]. There are three main arguments in the model.

- (i) We argue that after Pi release the binding energy of the ADP-head with the local MT-binding site (denoted by E_{W1}) is temporally weaker than that with other unperturbed MT-binding sites (denoted by $E_{W2} > E_{W1}$), which is understood as follows. The conformational changes in the local MT-tubulin heterodimer induced by strong binding of nucleotide-free, ATP- or ADP.Pi-head [32–37] result in the local tubulin having a further weaker interaction with the ADP-head than other unperturbed tubulins.¹ In a time of t_r the local tubulin relaxes

¹ With available structural data [37], our preliminary all-atom molecular dynamics simulations confirmed that the local MT tubulin, whose conformations are largely changed by strong interaction with the nucleotide-free kinesin head, has a much weaker binding affinity to ADP-head than unperturbed MT tubulins, with the former being roughly $12k_B T$ smaller than the latter, thus validating the argument.

to its normally unperturbed conformation, with the interaction energy of the local tubulin with the ADP-head becoming the same as other tubulins. Based on the argument, the interaction potential between the kinesin head and MT along a MT protofilament in an ATPase cycle is approximately shown in left panels of Fig. 1 (see Section S1 in Supplementary Information (SI) for detailed mathematical description of the interaction potential).

(ii) Based on the available experimental data [38,39], we argue

$$\Gamma_x \frac{\partial x}{\partial t} = -\frac{\partial V_W(x, y, z, \alpha, \theta)}{\partial x} - \frac{\partial V_I(x, y, z, \alpha, \theta)}{\partial x} - \frac{\partial V_{NL}(x)}{\partial x} - F_{NL} \left(\frac{r}{2} \right) \frac{x}{r} + \xi_x(t), \quad \text{when } |x - x_{bead}| < |x_{bead}|, \quad (1)$$

$$\Gamma_x \frac{\partial x}{\partial t} = -\frac{\partial V_W(x, y, z, \alpha, \theta)}{\partial x} - \frac{\partial V_I(x, y, z, \alpha, \theta)}{\partial x} - \frac{\partial V_{NL}(x)}{\partial x} - F_{NL} \left(\frac{r}{2} \right) \frac{x}{r} - C \left[(x - x_{bead}) - \text{sgn}(x - x_{bead}) \frac{d}{2} \right] H \left(|x - x_{bead}| - \frac{d}{2} \right) + \xi_x(t), \quad \text{when } |x - x_{bead}| \geq |x_{bead}|, \quad (2)$$

$$\Gamma_x^{(bead)} \frac{\partial x_{bead}}{\partial t} = C \left[(0 - x_{bead}) - \text{sgn}(0 - x_{bead}) \frac{d}{2} \right] H \left(|x_{bead}| - \frac{d}{2} \right) + F_x + \xi_x^{(bead)}(t), \quad \text{when } |x - x_{bead}| < |x_{bead}|, \quad (3)$$

$$\Gamma_x^{(bead)} \frac{\partial x_{bead}}{\partial t} = C \left[(x - x_{bead}) - \text{sgn}(x - x_{bead}) \frac{d}{2} \right] H \left(|x - x_{bead}| - \frac{d}{2} \right) + F_x + \xi_x^{(bead)}(t), \quad \text{when } |x - x_{bead}| \geq |x_{bead}|, \quad (4)$$

that when an MT-bound kinesin head is in ATP or ADP.Pi state there is a small free energy (denoted by E_{NL}) to facilitate its neck linker docking into the motor domain (see Section S2 in SI for detailed mathematical description of the potential of the effect of neck linker docking on the movement of other head), while when the head is in nucleotide-free or ADP state no such free energy is present to facilitate the neck linker docking.

(iii) We argue that there exists the interaction between the two heads, with the binding affinity between them depending on the nucleotide states of the two heads. Since during stepping the detached head is in ADP state and the MT-bound head can be in ATP or ADP.Pi or nucleotide-free state, we argue that when the MT-bound head is in nucleotide-free state the two heads have a high binding energy (denoted by E_{I1}), while when the MT-bound head is in ATP or ADP.Pi state the two heads have a low binding energy (denoted by $E_{I2} < E_{I1}$). In other words, the neck linker docking weakens the interaction between the two heads (see Section S3 in SI for detailed mathematical description of the potential of the interaction between the two kinesin heads). Possible origins of weakening of the interaction upon neck linker docking can be understood as follows. The neck linker docking results in a conformational change of the head, altering its interaction with the other head. Alternatively, the neck linker docking causes the change in the position of the N-terminus of the coiled-coil relative to the two heads, interfering the interaction between the two heads.

2.2. Equations to describe the movement of ADP-head relative to MT-bound head

Then, we present equations for the movement and rotation of the ADP-head relative to the nucleotide-free or ATP-head bound strongly to MT. We define the coordinate xyz as shown in Fig. 1a, where the origin of the coordinate (0,0,0) is at the center-of-mass position of the MT-bound head. For simplicity but without loss of generality, we do not consider here the precession motion of the head (i.e., the rotation around the direction of its neck linker), which corresponds to the rotation in the yz plane when the head is in the MT-binding state. Thus, we need to consider the movement of the ADP-head in three dimensions (denoted by coordinates x, y and z) and nutation (denoted by angle α) and rotation (denoted by angle

θ) motions (when the kinesin head is in the MT-binding site, α and θ correspond to the angles of rotations in xoz and xoy planes, respectively).

As done in the single-molecule optical trapping experiments [11–25,27], we consider a bead with diameter of $2R_{bead}$ attached to the coiled-coil stalk of the dimer and a longitudinal force (the force along or opposite to the x direction) acting on the bead. With one head bound strongly to MT at position (0,0,0), the motion and rotation of the other ADP-head relative to the MT-bound head in viscous solution can be described by Langevin equations [31]

$$\Gamma_y \frac{\partial y}{\partial t} = -\frac{\partial V_W(x, y, z, \alpha, \theta)}{\partial y} - \frac{\partial V_I(x, y, z, \alpha, \theta)}{\partial y} - F_{NL} \left(\frac{r}{2} \right) \frac{y}{r} + \xi_y(t), \quad (5)$$

$$\Gamma_z \frac{\partial z}{\partial t} = -\frac{\partial V_W(x, y, z, \alpha, \theta)}{\partial z} - \frac{\partial V_I(x, y, z, \alpha, \theta)}{\partial z} - F_{NL} \left(\frac{r}{2} \right) \frac{z}{r} + \xi_z(t), \quad (6)$$

$$\Gamma_\alpha \frac{\partial \alpha}{\partial t} = -\frac{\partial V_W(x, y, z, \alpha, \theta)}{\partial \alpha} - \frac{\partial V_I(x, y, z, \alpha, \theta)}{\partial \alpha} + \xi_\alpha(t), \quad (7)$$

$$\Gamma_\theta \frac{\partial \theta}{\partial t} = -\frac{\partial V_W(x, y, z, \alpha, \theta)}{\partial \theta} - \frac{\partial V_I(x, y, z, \alpha, \theta)}{\partial \theta} + \xi_\theta(t), \quad (8)$$

where x_{bead} denotes the x -coordinate of the center-of-mass position of the bead and $r = \sqrt{x^2 + y^2 + z^2}$. Due to the steric restriction of MT and considering the size of the kinesin head with radius $r_{head} = 2.5$ nm, it is required that $y \geq y_0 = 0$ and $r \geq 2r_{head} = 5$ nm. $V_W(x, y, z, \alpha, \theta)$ is the potential of the kinesin head interacting with MT-binding site during the period after Pi release and before ADP release (see Section S1 in SI), $V_I(x, y, z, \alpha, \theta)$ is the potential of interaction between two kinesin heads (see Section S3 in SI), $V_{NL}(x)$ is the potential characterizing the effect of the neck linker docking to the MT-bound head on the motion of the detached ADP-head (see Section S2 in SI), and F_{NL} is the force on the ADP-head that results from the stretched neck linkers (see Section S4 in SI). F_x is the longitudinal load acting on the bead, which is defined as negative when it points backward (i.e., the $-x$ direction) and positive when it points forward (i.e., the $+x$ direction), $\text{sgn}(x)$ is the sign function, and function $H(x)$ is defined as follows: $H(x) = 1$ if $x > 0$ and $H(x) = 0$ if $x \leq 0$. Here, it is considered that the interaction between the bead and the kinesin dimer acts only on the neck linker of the head that has a larger distance to the bead, and when the distance between a kinesin head and bead is smaller than $d/2$ no internally elastic force exists between them, with $d = 8$ nm being the distance between two successive binding sites along a MT protofilament. The drag coefficients on the kinesin head are $\Gamma_x = \Gamma_y = \Gamma_z = 6\pi\eta_0 r_{head}$ and $\Gamma_\alpha = \Gamma_\theta = 8\pi\eta_0 r_{head}^3$, where η_0 is the solution viscosity in the vicinity of MT. Since the viscosity in the vicinity of MT is larger than that far away from MT, we take $\eta_0 = 0.02$ g cm⁻¹ s⁻¹ that is about 2-fold larger than that in water. $\xi_i(t)$ ($i = x, y, z, \alpha, \theta$) is the fluctuating Langevin force on the kinesin head, with $\langle \xi_i(t) \rangle = 0$, $\langle \xi_i(t) \xi_j(t') \rangle = 0$ ($i \neq j$) and $\langle \xi_i(t) \xi_i(t') \rangle = 2k_B T \Gamma_i \delta(t - t')$, where $k_B T$ is the thermal energy. The drag coefficient on the bead is $\Gamma_x^{(bead)} = 6\pi\eta_0 R_{bead}$. $\xi_x^{(bead)}(t)$ is the fluctuating Langevin force on the bead, with $\langle \xi_x^{(bead)}(t) \rangle = 0$ and $\langle \xi_x^{(bead)}(t) \xi_x^{(bead)}(t') \rangle = 2k_B T \cdot \Gamma_x^{(bead)} \delta(t - t')$. The initial conditions

for Eqs. (1)–(8) are $(x_0, y_0, z_0, \alpha_0, \theta_0) = (-d, 0, 0, 0, 0)$. The connection between the bead and C-terminus ends of the neck linkers is characterized by an elastic linear spring with a spring constant C . In the calculation we take $C = 0.1$ pN/nm [31].

Using all-atom molecular dynamics (MD) simulations it was revealed that the coiled-coil stalk of kinesin-1 is a very stable structure, with the pulling force on the two neck linkers needed to unwind the coiled-coil being larger than the unbinding force of the nucleotide-free or ATP-head from MT [40]. Thus, it is considered that during the processive movement of kinesin-1, the coiled-coil stalk keeps its stable structure and it is not unwound or overwound. Single-molecule studies showed that during the processive movement, in the states with two heads bound simultaneously to MT the coiled-coil stalk is kept in the nearly fixed orientation [41,42]. Based on these pieces of evidence, we take the followings into account in our simulations of the limping dynamics. For odd steps, at the initial conditions with the two heads bound to MT the neck linker of each head is stretched to a length $r_i^{(0)} = d + \delta/2$ (see e.g., Fig. 1d) and after the binding to the forward MT-binding site the neck linker is stretched to a length $r_f^{(0)} = d - \delta/2$ (see, e.g., Fig. 1f); for even steps, at the initial conditions the neck linker is stretched to a length $r_i^{(E)} = d - \delta/2$ (see, e.g., Fig. 1f) and after the binding to the forward MT-binding site the neck linker is stretched to a length $r_f^{(E)} = d + \delta/2$ (see e.g., Fig. 1d). Here, δ ($\ll d$), which is called biased distance, is the distance between the C-terminus end of one neck linker and that of another neck linker along the x direction at the initial or final conditions when the two heads are bound to MT (see, e.g., Fig. 1d or f). In the simulations of other aspects of the movement dynamics without focusing on the limping effect, we take $\delta = 0$ for simplicity.

2.3. Monte-Carlo simulations of processive movement of dimeric kinesin

Using Eqs. (1)–(8) we can simulate the mechanical step of the movement of a kinesin head or bead following Pi release relative to the other MT-bound kinesin head by using stochastic Runge–Kutta algorithm, as done before [29,31]. Then, we can simulate processive movement of the dimer by also considering continuous ATPase activities, which can be simulated using Monte Carlo algorithm. In the Monte Carlo simulations, during each time step Δt ($\Delta t = 10^{-4}$ s in our simulation) a random number ran is generated with uniform probability between 0 and 1. The state transition with rate constant k_i ($i = T, c, -D$) takes place if $ran \leq P_i$ and the transition does not take place if $ran > P_i$. Here, $P_i = k_i \Delta t$ is the probability of state transition in each time step Δt , $k_T = k_b^{(T)}[ATP]$ represents ATP binding rate to the nucleotide-free head, with $k_b^{(T)}$ being the second-order rate constant for ATP binding and $[ATP]$ being the ATP concentration, k_c represents ATP-turnover rate (i.e., the rate of ATP hydrolysis and Pi release), and k_{-D} represents ADP-release rate from ADP-head. From each of our simulated 1000 traces of displacement versus time with a timescale of $\tau = 10$ –1000 s, the velocity is calculated from the total displacement divided by τ . The mean velocity is computed from all of 1000 individual velocities that are obtained from 1000 movement traces.

3. Results and discussion

3.1. The choice of parameter values

The choice of values of the parameters related to the interaction between the two kinesin heads, neck linker docking of the MT-bound head and interaction between the kinesin head and MT is discussed as follows. As done in our previous work [31], we

take $E_{I2} \leq 20k_B T$, $E_{I1} = 32k_B T$ and $t_r = 10$ μ s for kinesin-1 throughout. We also take $E_{NL} = 4.5k_B T$ and $E_{w2} = 35k_B T$, as done in the previous work [31], unless otherwise pointed out. Here, we take E_{w1} to be adjustable, and since by taking $E_{w1} = 22.3k_B T$ in the previous work [31] a lot of experimental data were explained quantitatively, the value of E_{w1} should be adjusted around $22.3k_B T$. For kinesin-2, values of above parameters will be given in Section 3.7.

The parameters related to ATPase activity are described as follows. As done in our previous work [31], we consider that the external load affects only the moving step in our model and the ATPase activity is independent of the load, which is consistent with the experimental data of Yildiz et al. [43] showing that extending the neck linker by any length (implying that the tension on the neck linker is varied) has no effect on ATPase activity. Moreover, we consider that when the head binds to the forward MT-binding site its ATP-turnover rate $k_c^{(lead)}$ is much smaller than ATP-turnover rate $k_c^{(trail)}$ ($k_c^{(trail)} \equiv k_c$) when it binds to the rear MT-binding site, i.e., when the neck linker is undocked the ATP-turnover rate is much smaller than that when the neck linker is docked, with $k_c^{(lead)} = k_c^{(trail)} / \rho = k_c / \rho$, where $\rho \gg 1$ is a dimensionless constant. Here, we fix $\rho = 40$ and take k_c to be adjustable. In the intermediate state, the MT-bound head with undocked neck linker also has the ATP-turnover rate equal to $k_c^{(lead)}$, while the MT-bound head with docked neck linker also has the ATP-turnover rate equal to $k_c^{(trail)}$. The second-order rate constant, k_b , for ATP binding always has the same value during the processive movement. Whether the ADP-head is in the rear MT-binding site or in the forward MT-binding site, the MT-stimulated ADP-release rate k_{-D} has the same non-zero value. When ADP-head is detached from MT, the ADP-release rate has a zero value. For kinesin-1, throughout we take $k_b = 2.5$ μ M $^{-1}$ s $^{-1}$, which is consistent with the biochemical data [44], $k_{-D} = 250$ s $^{-1}$ when ADP-head is bound to MT, which is consistent with the experimental data [4,45]. For kinesin-2, values of k_b , k_c and k_{-D} will be given in Section 3.7. Throughout we consider saturating concentration of ATP (2 mM). In Sections 3.2–3.6 we focus on kinesin-1, while in Section 3.7 we focus on kinesin-2.

3.2. Dynamics of limping stepping

To study the dynamics of limping stepping, we should also consider the vertical component of the external load acting on the bead attached to the coiled-coil (denoted by F_y) besides the longitudinal component (denoted by F_x), because F_y affects sensitively the limping effect [27]. It is evident that when the two heads are bound simultaneously to MT each head experiences a vertical force because the two heads have the same distance to the bead along the vertical y direction, and when only one head is bound to MT and the other head is detached from MT only the MT-bound head experiences a vertical force because the distance of the bead to the MT-bound kinesin head along the vertical y direction is larger than that to the detached kinesin head. For approximation, we consider here that when the ADP-head is in the range of $0 \leq y < 1$ nm (the MT is in the position of $y = 0$) the ADP-head experiences the vertical force, and when in other ranges ($y \geq 1$ nm) the ADP-head experiences no vertical force. Thus, under the consideration of both F_x and F_y acting on the bead, the motion and rotation of the ADP-head relative to the MT-bound head are described by Eqs. (1)–(8) but with Eq. (5) being replaced with following equations

$$\Gamma_y \frac{\partial y}{\partial t} = -\frac{\partial V_W(x, y, z, \alpha, \theta)}{\partial y} - \frac{\partial V_I(x, y, z, \alpha, \theta)}{\partial y} - F_{NL} \left(\frac{r}{2} \right) \frac{y}{r} + C(y_{bead} - y) + \xi_y(t), \quad \text{when } y < 1 \text{ nm}, \quad (9)$$

$$\Gamma_y \frac{\partial y}{\partial t} = -\frac{\partial V_W(x, y, z, \alpha, \theta)}{\partial y} - \frac{\partial V_I(x, y, z, \alpha, \theta)}{\partial y}, \quad (10)$$

$$-F_{NL} \left(\frac{r}{2} \right) \frac{y}{r} + \xi_y(t), \quad \text{when } y \geq 1 \text{ nm}$$

$$\Gamma_y^{(bead)} \frac{\partial y_{bead}}{\partial t} = C[(y - y_{bead}) + (0 - y_{bead})] + F_y + \xi_y^{(bead)}(t), \quad \text{when } y < 1 \text{ nm} \quad (11)$$

$$\Gamma_y^{(bead)} \frac{\partial y_{bead}}{\partial t} = C[(0 - y_{bead})] + F_y + \xi_y^{(bead)}(t), \quad (12)$$

$$\text{when } y \geq 1 \text{ nm},$$

where y_{bead} denotes the y -coordinate of the center-of-mass position of the bead. The drag coefficient on the bead along the y direction is $\Gamma_y^{(bead)} = 6\pi\eta_0 R_{bead}$. $\xi_y^{(bead)}(t)$ is the fluctuating Langevin force on the bead along the y direction, with $\langle \xi_y^{(bead)}(t) \rangle = 0$ and $\langle \xi_y^{(bead)}(t) \xi_y^{(bead)}(t') \rangle = 2k_B T \cdot \Gamma_y^{(bead)} \delta(t - t')$. In other words, we use Eqs. (1)–(4) and (6)–(12) to study the limping dynamics.

First, we consider the fixed longitudinal force, $F_x = -4$ pN, but varying vertical force F_y , as done in the experiments of Asbury et al. [21] and Fehr et al. [27]. Throughout this section we take $k_c = 112 \text{ s}^{-1}$. The sequence of the neck linker is taken from *Drosophila* kinesin, with the force-extension relation of the neck linker calculated with MD simulations (see Section S5 in SI) being shown in Fig. S1. In Fig. 2a, we show our calculated results of the mean dwell time of the odd steps and that of the even steps with $\delta = 0.18$ nm which is much smaller than or only about two percent of $d = 8$ nm. It is seen that the odd steps have a longer mean dwell time than the even steps, with the former called slow steps and the latter called the fast steps. More interestingly, it is seen that the mean dwell time of the slow steps increases with the increase of F_y , whereas the mean dwell time of the fast steps remains invariant. In Fig. 2b, we show the calculated results of the ratio of the mean dwell time of the slow steps to that of the fast steps by using results of Fig. 2a, with the ratio being also called limping factor. As it is seen, the limping factor increases with the increase of F_y . All these results are in agreement with the experimental data [21,27].

Then, we consider the variation of the longitudinal force F_x acting on the bead attached to the coiled-coil of kinesin Dmk401 used in the experiments of Fehr et al. [27], with the bead having a fixed radius R_{bead} . Accordingly, the vertical force F_y is calculated by $F_y = |F_x| \tan \Theta$ and $\sin \Theta = R_{bead} / (R_{bead} + l_{kinesin})$ [27], where $l_{kinesin}$ is the length of the coiled-coil stalk of Dmk401. As used in the experiments [27], we take $2R_{bead} = 0.44 \mu\text{m}$ and $l_{kinesin} = 21$ nm for Dmk401. In Fig. 3 we show our calculated results of the limping factor versus F_x with $\delta = 0.18$ nm. It is seen that the limping factor increases with the increase of F_x . These theoretical results are also in agreement with the experimental data [23,27].

The origin of limping and the feature that the limping factor increases with the increase of F_y is described as follows. Due to the existence of δ , it is expected that both the detaching process of the ADP-head from the previous MT-binding site and the binding process of the ADP-head to the next MT-binding site in the odd step can result in the mean dwell time being different from that in the even step. Our calculations show that with $E_{w1} = 22.3 k_B T$, the detaching probability in the odd step is about 0.82 and in the even step is about 0.97, giving the limping factor of only about 1.18. Thus, the large limping factor arises mainly from the binding process of the detached ADP-head to the next MT-binding site with the binding affinity of $E_{w2} = 35 k_B T$, which can be understood as follows. Under the action of the internal longitudinal force arising from the stretching of the neck linker, the longitudinal external force F_x and the external vertical force F_y , after binding to the next MT-binding

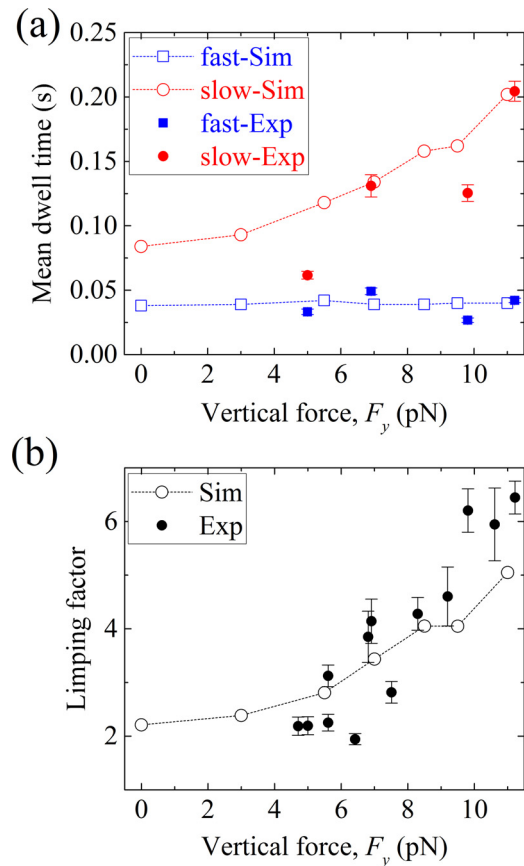


Fig. 2. Limping of kinesin-1 homodimer versus vertical force, F_y . The longitudinal force is fixed to $F_x = -4$ pN. The value of δ is taken to be 0.18 nm. (a) The mean dwell time of the slow (red circles) and fast stepping phases (blue squares) versus vertical force F_y . (b) Limping factor, i.e., the ratio of the mean dwell time of the slow stepping phase to that of the fast stepping phase calculated with the results of Fig. 2a. Unfilled symbols are theoretical data and filled symbols are experimental data taken from Asbury et al. [21] and Fehr et al. [27]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

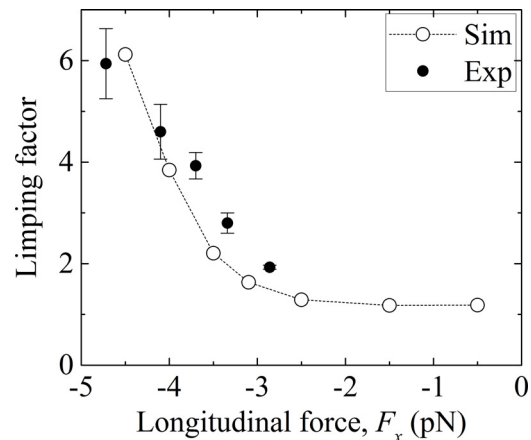


Fig. 3. Limping of kinesin-1 homodimer versus longitudinal force, F_x . The vertical force F_y is calculated by $F_y = |F_x| \tan \Theta$ and $\sin \Theta = R_{bead} / (R_{bead} + l_{kinesin})$, where R_{bead} is the radius of the bead attached to the coiled-coil stalk of the kinesin dimer and $l_{kinesin}$ is the stalk length. The value of δ is taken to be 0.18 nm. Unfilled symbols are theoretical data and filled symbols are experimental data taken from Fehr et al. [27].

site and before MT-stimulated ADP release the ADP-head can be re-detached from the next MT-binding site. Then, under the action of the backward external force F_x the re-detached ADP-head can re-diffuse to the previous MT-binding site whose binding affinity

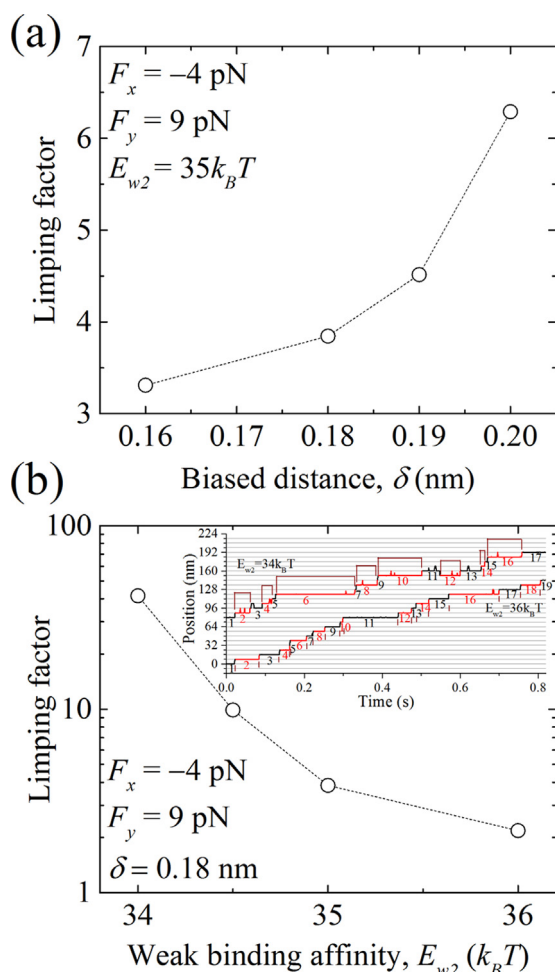


Fig. 4. Effects of biased distance δ and weak binding affinity E_{w2} on limping of kinesin-1 homodimer. The longitudinal force is fixed to $F_x = -4$ pN and the vertical force is fixed to $F_y = 9$ pN. (a) Limping factor versus δ for fixed $E_{w2} = 35k_B T$. (b) Limping factor versus E_{w2} for fixed $\delta = 0.18$ nm. Inset shows two typical movement traces with $E_{w2} = 34k_B T$ (upper trace) and $36k_B T$ (lower trace), separately.

has changed from $E_{w1} = 22.3k_B T$ to $E_{w2} = 35k_B T$, rebind to it and then ADP releases (noting $F_x = 0$ in the previous MT-binding site), giving a futile mechanochemical coupling cycle and thus greatly elongating the dwell time. Due to the existence of δ , the internal forces are different in odd and even steps, thus giving different probabilities of the ADP-head that has just bound to the next MT-binding site being re-detached from MT before ADP release and thus giving different mean dwell times. As the vertical external force F_y increases, the probability of the ADP-head that has just bound to the next MT-binding site being re-detached from MT before ADP release also increases, thus intensifying the difference between the mean dwell time of the odd steps and that of the even steps. Similarly, the feature that the increase of the external longitudinal force F_x increases the limping factor is easily understandable.

As discussed above, the limping factor is sensitive to two parameters δ and E_{w2} . Up to now (Figs. 2 and 3), we have fixed $\delta = 0.18$ nm and $E_{w2} = 35k_B T$. To see the effect of δ and E_{w2} on the limping, we calculate limping factor versus δ for fixed $E_{w2} = 35k_B T$ and limping factor versus E_{w2} for fixed $\delta = 0.18$ nm, with the results being shown in Fig. 4a and b, respectively. As expected, for the fixed E_{w2} , the limping factor increases significantly with the increase of δ (Fig. 4a); and for the fixed δ , the limping factor increases significantly with the decrease of E_{w2} (Fig. 4b), which can also be seen from inset of Fig. 4b where we show two typical movement traces with $E_{w2} = 34k_B T$ and $36k_B T$, separately. With the results of Fig. 4b

the experimental data of Higuchi et al. [23] can be explained as follows. The experimental data showed that under a given fixed longitudinal force F_x , the kinesin-T94S mutant shows an evidently larger limping factor than the wild-type kinesin, especially at large values of F_x [23]. With our results of Fig. 4b, this can be explained by considering that the mutation causes a small structural change of the motor domain, which slightly alters the binding site of the motor domain to MT, thus slightly reducing the binding affinity of ADP-head to MT, E_{w2} . As it is seen from Fig. 4b, by reducing E_{w2} from $36k_B T$ to $34.5k_B T$ (reducing by only about 4.2%), the limping factor increases from 2.18 to 9.91 (increases by about 4.5-fold).

3.3. Effects of external longitudinal force on velocity

As mentioned in Introduction, many research groups have measured the movement velocity of the kinesin-1 dimer versus both backward and forward longitudinal loads acting on the bead attached to the coiled-coil stalk. Puzzlingly, some experimental data showed that the velocity versus the longitudinal load has a sigmoid form, i.e., the velocity decreases with the increase of the backward load whereas increases little or does not increase with the increase of the forward load [20], while other data showed that the velocity versus external load has a double-sigmoid form, i.e., the velocity decreases with the increase of the backward load and increases with the increase of the forward load [13,24]. In this section, we make a consistent explanation of these contradictory experimental data. For simplicity of analysis, we neglect the effect of the vertical component of the external force, i.e., take $F_y = 0$ or use Eqs. (1)–(8), and additionally we take $\delta = 0$ in this section and the following sections, where the limping effect is not considered, as done in the previous work [31].

It is reasonable to argue that different protein species or buffer conditions or protein preparations by different research groups can give different values of parameters E_{w1} and k_c . Thus, we take different values of E_{w1} and k_c to explain different experimental data. Here, we adjust the value of k_c to make the maximum velocity that the motor can attain under a large forward load be consistent with the available experimental data. Then, with the value of k_c we adjust the value of E_{w1} to make the calculated velocity under no load be consistent with the available experimental data. Values of other parameters are given in Section 3.1. The force-extension relation of the neck linker is given in Fig. S1. First, we focus on the experimental data of Coppin et al. [13] using native squid kinesin. We take $k_c = 34.4$ s $^{-1}$ and $E_{w1} = 23.2k_B T$ that is slightly larger than $22.3k_B T$ (see Section 3.1). In Fig. 5a (left panel) we show our calculated results of the movement velocity versus external longitudinal force F_x , and compare with the experimental data of Coppin et al. [13] (right panel). It is seen that the theoretical results are in agreement with the experimental data, showing that the velocity versus F_x has a double-sigmoid form.

Then, we focus on the experimental data of Carter and Cross [24] using *Drosophila* kinesin. We take $k_c = 108$ s $^{-1}$ and $E_{w1} = 23.5k_B T$ that is also slightly larger than $22.3k_B T$. The theoretical results of the velocity versus F_x (Fig. 5b, left panel) are also in agreement with the experimental data [24] (Fig. 5b, right panel), which also show a double-sigmoid form of velocity versus F_x .

Third, we focus on the experimental data of Block et al. [20] using kinesin purified from *Loligo pealei*. We take $k_c = 100$ s $^{-1}$ and $E_{w1} = 21.3k_B T$ that is slightly smaller than $22.3k_B T$. Furthermore, our calculations show that the results of velocity versus F_x are insensitive to E_{w1} provided that $E_{w1} \leq 21.3k_B T$. In Fig. 5c (left panel) we show that calculated results of the velocity versus F_x with $E_{w1} = 21.3k_B T$ and 0, separately. It is seen that the theoretical results are in agreement with the experimental data of Block et al. [20] (right panel, Fig. 5c), showing that the velocity does not increase with the positive value of F_x .

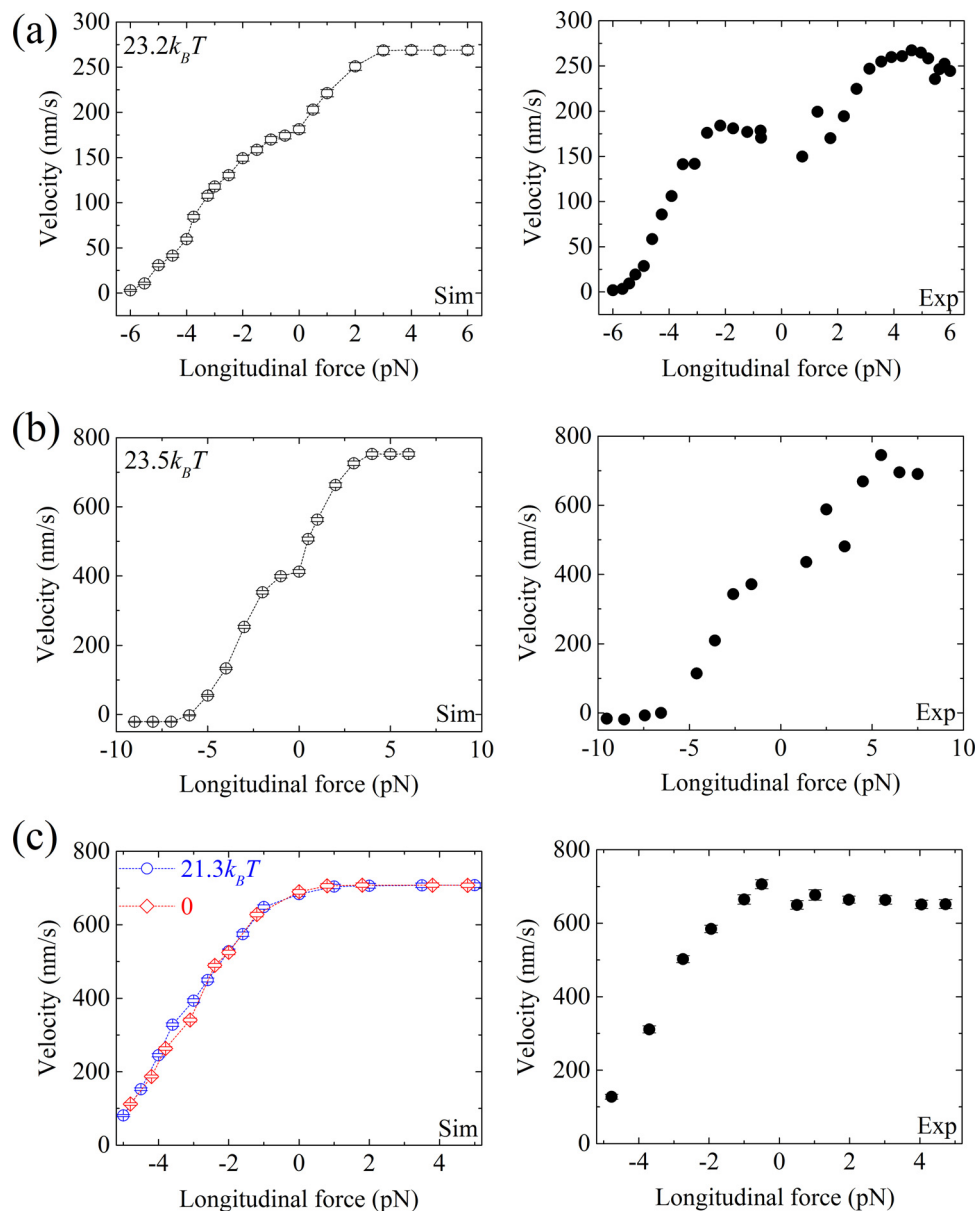


Fig. 5. Effects of external longitudinal force on velocity of kinesin-1. (a) Velocity versus forward and backward longitudinal force calculated with $k_c = 34.4 \text{ s}^{-1}$ and $E_{w1} = 23.2 k_B T$ (left panel). Right panels correspond to the experimental data taken from Coppin et al. [13]. (b) Velocity versus forward and backward longitudinal force calculated with $k_c = 108 \text{ s}^{-1}$ and $E_{w1} = 23.5 k_B T$ (left panel). Right panels correspond to the experimental data taken from Carter and Cross [24]. (c) Velocity versus forward and backward longitudinal force calculated with $k_c = 100 \text{ s}^{-1}$ and $E_{w1} \leq 21.3 k_B T$ (left panel). Right panels correspond to the experimental data taken from Block et al. [20].

The origins that in some cases the velocity makes further increase with the increase of the forward load while in other cases the velocity does not increase with the increase of the forward load are explained as follows. In our model, a forward stepping in an ATPase cycle requires the trailing ADP-head to escape the potential well of depth E_{w1} within time t_r , since after t_r the depth becomes $E_{w2} > E_{w1}$ so that the head is unable to escape the potential well before ADP is released. For cases such as Coppin et al. [13] and Carter and Cross [24], E_{w1} has larger values, giving the probability (P_{escape}) of the trailing ADP-head to escape the potential well of depth E_{w1} within time t_r evidently smaller than one under no external load. A forward load can increase the probability P_{escape} , resulting in the velocity to be larger than that under no external load. By contrast, for other cases such as Block et al. [20], E_{w1} has smaller values, giving the probability P_{escape} nearly equal to one under no external load. Thus, a forward load cannot make a further increase of

the probability P_{escape} , and thus velocity does not increase with the forward load.

3.4. Effects of extended neck linkers on velocity

In this section, we focus on studies of the effects of extended neck linkers on movement velocity. To this end, we consider a series of kinesin-1 constructs containing neck linkers that are extended incrementally, from 1 to 6 additional amino acids (AA) at the junction of the neck-linker domain and the coiled-coil stalk, as done in Andreasson et al. [46]. For convenience, we denote these kinesin-1 constructs by K-WT (kinesin-1 with no additional AA), K-1AA (kinesin-1 with one additional AA), ..., and K-6AA (kinesin-1 with six additional AA). Because the additional amino acids do not involve neck linker docking, it is expected that extending the neck linkers has no effect on the neck linker docking, interaction of the motor domain with the

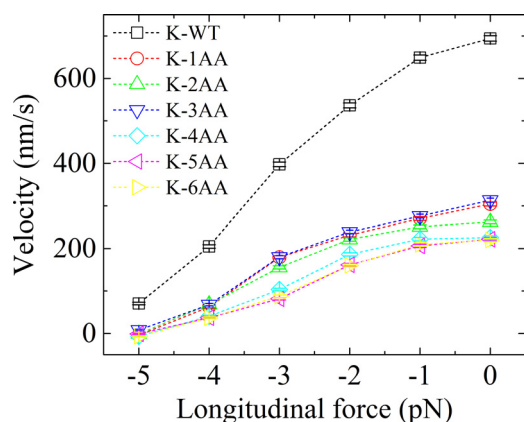


Fig. 6. Velocity versus backward longitudinal force for kinesin-1 dimers with extended neck linkers. K-WT represents kinesin-1 with no additional amino acids, K-1AA, K-2AA, K-3AA, K-4AA, K-5AA and K-6AA represent kinesin-1 with one, two, three, four, five and six additional amino acids, respectively, inserted at the junction of the neck-linker domain and the coiled-coil stalk.

MT-binding site, ATPase activity and interaction between the two heads. Thus, these kinesin constructs have the same values of all parameters except that their neck linkers have different lengths, with $E_{12} \leq 20k_B T$, $E_{11} = 32k_B T$, $t_r = 10 \mu s$, $E_{NL} = 4.5k_B T$, $E_{w2} = 35k_B T$, $E_{w1} = 22.3k_B T$, $k_b = 2.5 \mu M^{-1} \cdot s^{-1}$, $k_D = 250 s^{-1}$ (see Section 3.1) and $k_c = 112 s^{-1}$. For K-WT, the sequence of the neck linker is taken from *Drosophila* kinesin, with the force-extension relation of the neck linker being shown in Fig. S1. The sequences of the inserted amino acids in the lengthened neck linkers are as follows [46]: extended by 1 AA (L; K-1AA), 2 AA (HV; K-2AA), 3 AA (DAL; K-3AA), 4 AA (LAST; K-4AA), 5 AA (LASQT; K-5AA), and 6 AA (LQASQT; K-6AA). The force-extension relations of these lengthened neck linkers calculated with MD simulations are shown in Fig. S2.

In Fig. 6, we show our calculated results of the movement velocity versus external longitudinal force F_x for the series of kinesin constructs with extended neck linkers, and compare with the corresponding results for the kinesin with no AA inserted. Interestingly, it is seen that the insertion of a single AA in the neck linker causes a large drop in velocity across all loads and additional insertions produce only small changes in velocity. These are in agreement with the experimental data of Andreasson et al. [46].

3.5. Effects of extended neck linkers on half-site ADP release

In this section, we focus on studies of the effects of extended neck linkers on half-site ADP release, as done in the experiments of Andreasson et al. [46]. In the experiments, the mantADP-bound kinesin-1 dimers were mixed with MTs, and about half amount of the mantADPs bound to the dimers were released rapidly; then, the other half amount of the mantADPs were released very slowly. To see the effects of extended neck linkers, a series of truncated *Drosophila* kinesin-1 constructs (DmK) containing neck linkers that were extended incrementally, from 1 to 6 AA (DmK-1AA to DmK-6AA), were examined.

To simulate these experimental data of half-site mantADP-releasing rate, we consider that one head in nucleotide-free state is bound fixedly to MT and the other ADP-head is initially in the intermediate position where the two heads are bound together with an affinity of E_{11} . Once the ADP-head is deviated by the thermal noise from the intermediate position and then diffuses to either the next or the previous MT-binding site with a binding affinity of E_{w2} , the ADP will be released with a rate constant k_D . If the ADP-head bound to the MT-binding site is re-detached from the MT-binding site, ADP-releasing rate becomes zero. After ADP is released, the two nucleotide-free heads become bound simultaneously and strongly

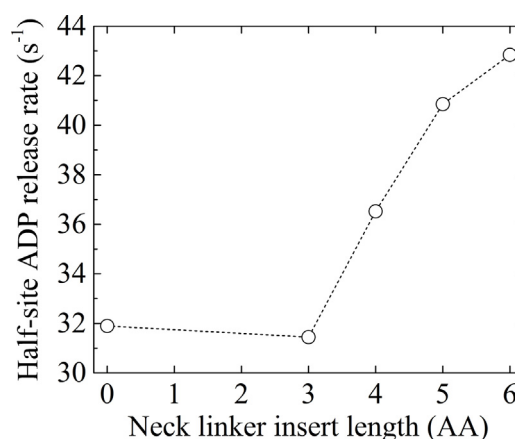


Fig. 7. The half-site ADP release rate from kinesin-1 dimers versus number of amino acids inserted at the junction of the neck-linker domain and the coiled-coil stalk.

to MT. With values of parameters as given in above Section 3.4, we calculate the rate of ADP releasing from the ADP-head (i.e., the inverse of the time for ADP to release from the ADP-head) for K-WT, K-1AA, ..., and K-6AA. In Fig. 7 we show our calculated results of the half-site ADP-releasing rate versus neck-linker insert length. It is seen that the constructs with neck linkers extended by up to 3 AA show nearly the same half-site ADP-releasing rate as the wild-type one, but that extending the neck linker by 4 AA or more leads to an abrupt increase in the half-site ADP-releasing rate. These features are consistent with the experimental data of Andreasson et al. [46].

3.6. Load-dependent velocity for different species and mutations in the neck linkers

Recently, using single-molecule optical trapings Andreasson et al. [46] made detailed studies and comparisons of the movement dynamics among two species of kinesin-1 constructs, their neck-linker mutants of elongated length, and cysteine-light (CL) mutants, where a residue cysteine in the neck linker domain was mutated. One species include wild-type truncated *Drosophila* kinesin construct (DmK-WT), truncated *Drosophila* kinesin construct with its neck linker being extended by inserting six additional amino acid residues into the C-terminal portion of the linker region of each head (DmK-6AA), and truncated cysteine-light (CL) *Drosophila* kinesin construct (DmK-CL). Another species include wild-type truncated human construct (HsK-WT), human construct with its neck linker being extended by inserting six additional amino acid residues into the C-terminal portion of the linker region of each head (HsK-6AA), and human CL constructs with no additional amino acid inserted into its neck linker (HsK-CL) and with six additional amino acid residues inserted into the C-terminal portion of the linker region of each head (HsK-CL-6AA). Intriguingly, they found that these kinesin constructs show very different features in the movement dynamics, as summarized as follows. (i) The force-velocity relation for DmK-6AA differs significantly from the corresponding relation for HsK-CL-6AA. Specifically, DmK-6AA exhibits a substantially greater velocity—which is also less force dependent—than HsK-CL-6AA. Whereas HsK-CL-6AA is capable of undertaking many rearward steps when subjected to hindering loads beyond the stall force, no such processive backstepping was observed for DmK-6AA, nor for any other construct. (ii) DmK-WT and HsK-WT exhibit nearly identical force-velocity relations under identical conditions. (iii) Under backward loads, DmK-CL and HsK-CL exhibit smaller velocities than DmK-WT and HsK-WT. Under forward loads, the velocities of DmK-CL and HsK-CL increase relative to those with no external load, whereas the velocities of

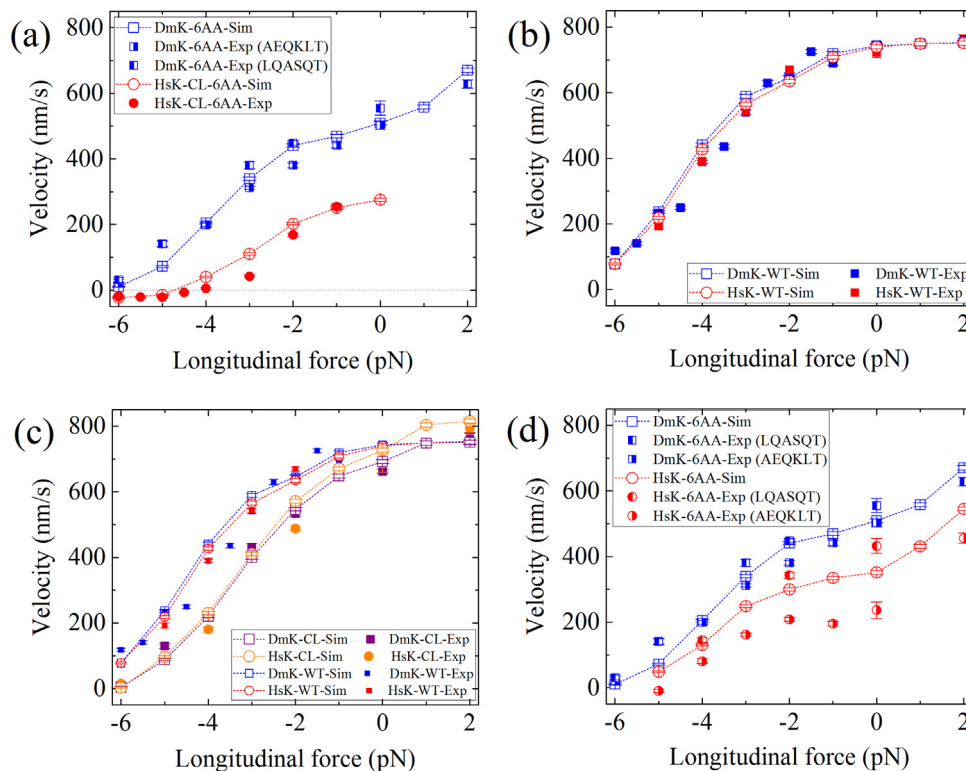


Fig. 8. Load-dependent velocity for different species of kinesin-1 and mutations in the neck linkers. Unfilled symbols are theoretical data, while half-filled and fully-filled symbols are experimental data taken from Andreasson et al. [46], with DmK-6AA-Exp(AEQKLT) and DmK-6AA-Exp(LQASQT) representing the experimental data for DmK-6AA with sequences of the six inserted amino acids being AEQKLT and LQASQT, respectively, and with HsK-6AA-Exp(AEQKLT) and HsK-6AA-Exp(LQASQT) representing the experimental data for HsK-6AA with sequences of the six inserted amino acids being AEQKLT and LQASQT, respectively. (a) Velocity versus longitudinal force for DmK-6AA and HsK-CL-6AA. (b) Velocity versus longitudinal force for DmK-WT and HsK-WT. (c) Velocity versus longitudinal force for DmK-CL and HsK-CL. For comparison, the corresponding data for DmK-WT and HsK-WT are also shown. (d) Velocity versus longitudinal force for DmK-6AA and HsK-6AA.

DmK-WT and HsK-WT do not increase. (iv) DmK-6AA exhibits larger velocity than HsK-6AA under the same backward load.

In this section, we make quantitative explanations of these intriguing experimental data. Since after docking the neck linker interacts with the motor domain [47], the mutations of residues in the neck linker domain that involve neck linker docking can affect slightly the interaction of the motor domain with the MT-binding site and ATPase activity besides the neck-linker-docking energy E_{NL} . By contrast, for the mutated dimer with the extension of its neck linker via inserting additional amino acid residues into the C-terminal portion of the linker region of each head, the inserted residues do not involve neck linker docking. Thus, extending the neck linkers would have no effect on the interaction of the motor domain with the MT-binding site, ATPase activity, neck-linker-docking energy and interaction between the two heads. As a result, it is expected that CL constructs of human kinesin, HsK-CL and HsK-CL-6AA, have the same values of all parameters except that their neck linkers have different lengths; HsK-CL and HsK-CL-6AA have values of E_{W1} , k_c and E_{NL} slightly different from those of HsK-WT and HsK-6AA. Similarly, CL constructs of *Drosophila* kinesin, DmK-CL and DmK-CL-6AA, have the same values of all the parameters except their neck linkers have different lengths, and DmK-CL and DmK-CL-6AA have values of E_{W1} , k_c and E_{NL} slightly different from DmK-WT and DmK-6AA. In our calculations, we take $E_{W1} = 22.1k_B T$, $k_c = 120 s^{-1}$ and $E_{NL} = 4.5k_B T$ for HsK-CL and HsK-CL-6AA, and $E_{W1} = 21.3k_B T$, $k_c = 108 s^{-1}$ and $E_{NL} = 6k_B T$ for HsK-WT and HsK-6AA, while all other parameters except those related to the force-extension relation of the neck linker have the same values (given in Section 3.1). Similarly, we take $E_{W1} = 21.8k_B T$, $k_c = 108 s^{-1}$ and $E_{NL} = 4.5k_B T$ for DmK-CL, and $E_{W1} = 20.5k_B T$, $k_c = 108 s^{-1}$ and $E_{NL} = 6k_B T$ for DmK-WT and DmK-6AA, while all other parameters

except those related to the force-extension relation of the neck linker have the same values (given in Section 3.1). The force-extension relations of the different neck linkers are calculated with MD simulations (Fig. S3).

In Fig. 8a, we show our calculated results of the velocity versus external load F_x for DmK-6AA and HsK-CL-6AA, and compare with the corresponding experimental data [46]. It is mentioned here that our calculations show that different sequences of the six inserted amino acids (LQASQT and AEQKLT) give very small difference in the velocity under the same load. From Fig. 8a it is seen that the simulation data are in good agreement with the experimental data. In particular, both simulation and experimental data show that DmK-6AA exhibits a substantially greater velocity than HsK-CL-6AA; whereas HsK-CL-6AA is capable of undertaking many rearward steps under $-6 pN \leq F_x \leq -5 pN$, no such processive backstepping is observed for DmK-6AA under $F_x \geq -6 pN$. In Fig. 8b, we show our calculated results of the velocity versus external load F_x for DmK-WT and HsK-WT, and compare with the corresponding experimental data [46]. It is seen that the simulation data are also in good agreement with the experimental data. Both simulation and experimental data show that DmK-WT and HsK-WT exhibit nearly identical force-velocity relations. In Fig. 8c, we show our calculated results of the velocity versus external load F_x for DmK-CL and HsK-CL, and compare with the corresponding experimental data [46]. For comparison, the corresponding data for DmK-WT and HsK-WT (shown in Fig. 8b) are also shown. It is seen that the simulation data are also in good agreement with the experimental data. In particular, Both simulation and experimental data show that under backward loads, DmK-CL and HsK-CL exhibit smaller velocities than DmK-WT and HsK-WT; under forward loads, the velocities of DmK-CL and HsK-CL increase relative to those with no external load,

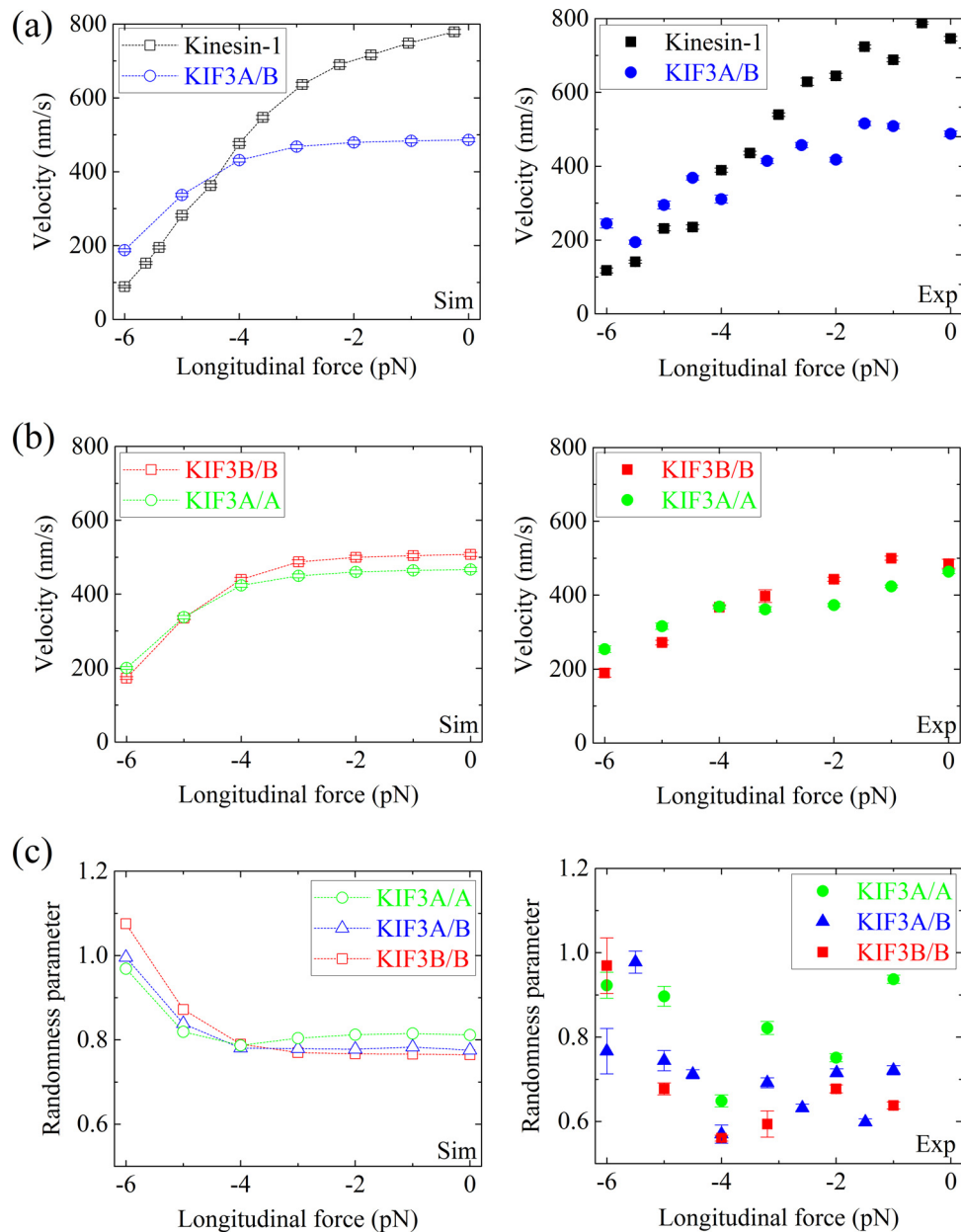


Fig. 9. Velocity and randomness parameter versus backward longitudinal force for kinesin-2. Left panels are theoretical data and right panels are experimental data taken from Andreasson et al. [28]. (a) Velocity versus backward longitudinal force for KIF3A/B. For comparison, the corresponding results for kinesin-1 are also shown. (b) Velocity versus backward longitudinal force for KIF3A/A and KIF3B/B. (c) Randomness parameter versus backward longitudinal force for KIF3A/A, KIF3A/B and KIF3B/B.

whereas the velocities of DmK-WT and Hsk-WT do not increase. In Fig. 8d, we show our calculated results of the velocity versus external load F_x for DmK-6AA and HsK-6AA, and compare with the corresponding experimental data [46]. The simulation data show that DmK-6AA exhibits larger velocity than HsK-6AA under the same backward load, which are consistent with the experimental data. Moreover, from Fig. 8a – d, it is seen that only HsK-CL-6AA is capable of undertaking many rearward steps when subjected to backward loads beyond the stall force, no any other construct shows such feature, which is also in agreement with the experimental data. Taken together, the intriguing experimental data for two species of kinesin-1 constructs, their neck-linker mutants of elongated length and cysteine-light mutants observed by Andreasson et al. [46] are explained well and quantitatively by our theoretical results.

3.7. Load-dependent velocity of kinesin-2

Up to now, we have focused on the studies of conventional kinesin—kinesin-1. Recent single-molecule data showed that kinesin-2 motors—heterodimeric KIF3A/B and homodimeric mutants KIF3A/A and KIF3B/B—have different load-dependent stepping dynamics from kinesin-1 [28]. Here, it is considered that both kinesin-1 and kinesin-2 share the same mechanism of processive stepping and the difference in the stepping dynamics between the two kinesin families arises from that the two have different values of some fundamental parameters.

Here, we assume that in nucleotide-free or ATP-state kinesin-2 has a larger binding affinity to MT than kinesin-1, implying that the conformational changes in the local MT-tubulin heterodimer induced by strong binding of nucleotide-free or ATP-head for kinesin-2 are larger than and very different from those for kinesin-

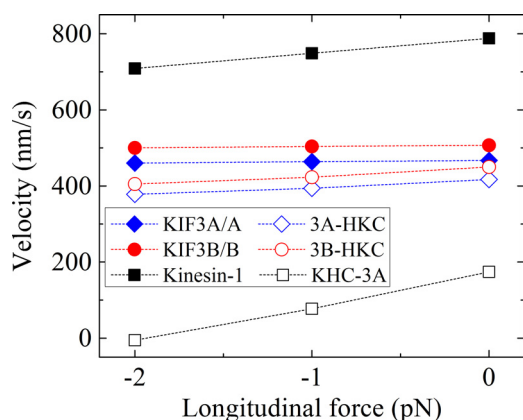


Fig. 10. Velocity of kinesin-1 and kinesin-2 with swapped neck linkers versus backward longitudinal force. 3A-KHC is the construct with KIF3A head and kinesin-1 neck linker, 3B-KHC is the construct with KIF3B head and kinesin-1 neck linker, and KHC-3A is the construct with kinesin-1 head and KIF3A neck linker.

1. Thus, it is expected that E_{w1} has a smaller value for kinesin-2 than that for kinesin-1 and the relaxation time t_r of the changed conformations of local MT-tubulin to the normal conformation for kinesin-2 is longer than or very different from that for kinesin-1. Additionally, different motor domains should have different values of ATP-turnover rate k_c . As a result, except E_{w1} , t_r , k_c and parameters related to the force-extension relations of the neck linkers, for simplicity we take all other parameters having the same values for kinesin-1 and kinesin-2, i.e., $E_{NL} = 6k_B T$, $E_{w2} = 35k_B T$, $E_{l2} \leq 20k_B T$, $k_b = 2.5 \mu\text{M}^{-1} \text{s}^{-1}$, $k_{-D} = 250 \text{s}^{-1}$ and $\rho = 40$, as done above (since we consider here saturating ATP concentration, the value of E_{l1} is not required). For kinesin-1, we take $k_c = 134 \text{s}^{-1}$, $E_{w1} = 22.3k_B T$ and $t_r = 10 \mu\text{s}$, as done above. For KIF3A, we take $k_c = 62 \text{s}^{-1}$, $E_{w1} \leq 20k_B T$ and $t_r = 3000 \mu\text{s}$; and for KIF3B, we take $k_c = 68 \text{s}^{-1}$, $E_{w1} \leq 20k_B T$ and $t_r = 2000 \mu\text{s}$. The force-extension relation of the neck linkers for kinesin-1, KIF3A/A and KIF3B/B are obtained from MD simulations (see Fig. S1 and Fig. S4).

In Fig. 9a (left panel) we show our calculated results of the movement velocity versus external load F_x for KIF3A/B, and compare with the corresponding single-molecule data [28] (right panel). For comparison, in Fig. 9a (left panel) we also show the corresponding results for kinesin-1. In Fig. 9b (left panel) we show the calculated results of the velocity versus external load F_x for KIF3A/A and KIF3B/B, and compare with the corresponding single-molecule data [28] (right panel). Fig. 9c (left panel) shows the calculated results of randomness parameter, RP , versus external load F_x for KIF3A/B, KIF3A/A and KIF3B/B, and the corresponding single-molecule data are shown in the right panel of Fig. 9c, where the randomness parameter RP is calculated by $RP = \lim_{t \rightarrow \infty} \left\{ \frac{[\langle x^2(t) \rangle - \langle x(t) \rangle^2]}{[d\langle x(t) \rangle]} \right\}$, as done before [16,28]. It is seen that all our theoretical data are in good agreement with the single-molecule data.

To see the effects of neck linker docking on movement dynamics of kinesin-2 and kinesin-1, we swap the neck-linker domains between kinesin-1 and kinesin-2, as done in Andreasson et al. [28]. Because the neck linker docking involves interactions with complementary residues in the core motor domain, swapping the neck linkers would destroy the neck linker docking, giving $E_{NL} = 0$, as mentioned in Andreasson et al. [28]. With $E_{NL} = 0$ and values of other parameters as given above, we calculate the velocities for 3A-KHC (with KIF3A head and kinesin-1 neck linker), for 3B-KHC (with KIF3B head and kinesin-1 neck linker) and for KHC-3A (with kinesin-1 head and KIF3A neck linker). The calculated results of the velocity versus F_x are shown in Fig. 10, where for comparison,

we also show the corresponding results for KIF3A/A, KIF3B/B and kinesin 1. It is seen that the results for 3A-KHC and 3B-KHC with kinesin-2 head and kinesin-1 neck linker behave similar to those for KIF3A/A and KIF3B/B; by contrast, KHC-3A with kinesin-1 head and kinesin-2 neck linker shows much slower velocities than kinesin-1. Thus, our results give quantitative explanations of the puzzling single-molecule data showing that “specific docking of the neck linker to the core motor domain is important for kinesin-1 but not for kinesin-2” [28].

4. Concluding remarks

In this work, using our model we study the limping dynamics of the kinesin-1 homodimer, providing quantitative explanations of the available single-molecule experimental data. The alternation in the mean dwell time in successive steps results from the alternation in the mechanochemical coupling efficiency, which is in turn caused by the alternation in the slight variation of the stretched neck linker length, while the ATPase activity is kept the same. Moreover, the conflicting single-molecule data, with some showing that the velocity of kinesin-1 does not change with the forward longitudinal load while others showing that the velocity increases largely with the forward longitudinal load, are also explained quantitatively and consistently. Additionally, the biochemical data on half-site ADP release for wild-type and mutant kinesin-1 constructs with extended neck linkers are also explained; the intriguing experimental data on different features of the movement dynamics among two species of kinesin-1 constructs (including *Drosophila* and human constructs), their neck-linker mutants of elongated length, and cysteine-light mutants are explained quantitatively; the puzzling experimental results showing that the specific docking of the neck linker to the core motor domain is important for kinesin-1 but not for kinesin-2 are also explained quantitatively and well.

In the previous work [31], with the same model a lot of experimental data on diverse aspects of movement dynamics of kinesin-1, such as the effects of longitudinal load acting on the bead attached to the coiled-coil stalk and on the bead attached to one head of the dimer, nullifying the neck linker docking, extending the neck linkers of cysteine-light kinesin-1 mutants, etc., on the movement velocity, mechanochemical coupling ratio, forward and backward stepping ratio, stall force and randomness parameter, were explained quantitatively and consistently. Particularly, the puzzling experimental data showing that the increase in solution viscosity, which has no effect on ATPase activity of kinesin-1, reduces significantly the movement velocity [48] were explained well and quantitatively [31].

A pivot in our model is the argument that after Pi release from a kinesin head, the locally changed conformation of the MT-binding site, which is induced by the strong interaction with the head in nucleotide-free, ATP and ADP.Pi states, relaxes to its normally unperturbed conformation with a delayed time relative to the Pi-release-induced conformational change of the head, thus resulting in the interaction energy of the ADP-head with the local MT-binding site to be temporally weaker than that with other unperturbed MT-binding sites.¹ With this argument, the single-molecule experimental data on processive movement of the monomeric kinesin KIF1A were also explained quantitatively [49]. Thus, various experimental data on the moving dynamics of both kinesin monomers and dimers can be explained consistently.

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in the simulations in the work can be found in github (<https://github.com/think92/Dynamics-of-dimeric-kinesins>) or provided upon requests.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ijbiomac.2017.07.147>.

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